

AKANSHA AGGARWAL

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Personal Statement

Computer Systems Engineering graduate with hands-on experience in robotics, embedded systems, AI/ML, data tracking, hardware troubleshooting, and technical documentation. Experienced in supporting real-world technical deployments, analysing system data, producing operational records, and collaborating with teams to improve workflows. Strong interest in robotics operations, AI-driven workflows, dashboards, automation, and field technology.

Education

Bachelor of Engineering (Honours), Computer Systems

University of Auckland | Feb 2022 – Dec 2025

- Grade: Second Class Honours (First Division)
- Relevant coursework: Embedded Systems, Software Engineering, Artificial Intelligence, Machine Learning, Computer Architecture

Technical Skills

Robotics & Hardware Systems: Embedded systems, microcontrollers, FPGA-based systems, robotics hardware integration, sensor-based systems, hardware troubleshooting, real-time systems, system monitoring

AI, Data & Operations: AI/ML workflows, data processing, data validation, operational reporting, system logging, workflow optimisation, data tracking, record accuracy, anomaly detection concepts

Programming & Automation: Python, Pandas, Java, C, C++, JavaScript, SQL, REST APIs, scripting, automation concepts, backend integration

Tools & Platforms: Git, Visual Studio, MATLAB, Quartus II, Altium Designer, Excel, ERP concepts, asset tracking systems, version control systems

Operations & Documentation: Technical documentation, operational reporting, process compliance, troubleshooting, inventory and asset tracking, serial number tracking, auditing support, knowledge base maintenance

Collaboration & Workflow Management: Cross-functional collaboration, customer support, task coordination, problem-solving, workflow management, communication across technical and non-technical teams

Projects

Smart Energy Monitor for Racing Cars

- Collected, processed, and analysed real-time telemetry and energy monitoring data from embedded systems
- Integrated hardware and software systems for reliable data capture and system performance
- Logged and analysed system outputs for monitoring and troubleshooting

Face Recognition System

- Developed computer vision and deep learning models (VGG, ResNet) for face recognition and image-based classification
- Trained and evaluated models using benchmark datasets to assess accuracy and performance
- Applied optimisation techniques to improve model efficiency and reliability
- Applied fundamental testing concepts to design test cases and validate system functionality throughout the development lifecycle

Industrial Automation & Step Detection System

- Developed an ABS conveyor control system using SystemJ for synchronous reactive modelling
- Implemented motion step detection using sensor orientation and angular analysis in Java
- Demonstrated strong understanding of real-time systems and software-hardware interaction
- Performed testing and debugging to ensure system reliability and performance

Pacemaker System

- Developed a DDD-mode pacemaker using SCCharts and C on FPGA
- Integrated UART communication for testing with a virtual heart simulation
- Focused on reliability and safety-critical system design

Work Experience

Research Assistant – University of Auckland | Jun 2025 – Feb 2026

- Installed, configured, and supported technical systems for job tracking and power monitoring
- Handled system data and tracking records, ensuring accuracy, validation, and traceability
- Assisted in troubleshooting hardware and software issues across multiple deployments
- Maintained operational documentation and structured system records
- Worked with real-world datasets to ensure system reliability and data integrity
- Collaborated with teams to improve workflows and support system operations

Customer Service Representative – Noel Leeming (Auckland) | Oct 2024 – Mar 2025

- Processed transactions and handled customer requests with high accuracy
- Managed product handling and stock interactions in a fast-paced environment
- Developed strong attention to detail and communication skills

UniCrew – University of Auckland | Dec 2024 – Dec 2025

- Supported university events, logistics, and student engagement initiatives
- Guided new students in navigating campus services and resources
- Fostered an inclusive and supportive environment through effective communication

Junior Web Developer – Beyond Studio (India) | Jun 2023 – Dec 2024

- Developed responsive websites using WordPress and Wix
- Built and integrated REST APIs for data-driven applications
- Collaborated with cross-functional teams to deliver projects and improve user experience
- Assisted in testing, troubleshooting, and optimising website functionality and performance

Research Experience

Companion Robot Design (Research Project)

Supervisor: Professor Bruce MacDonald

- Designed and developed an AI-assisted socially interactive companion robot focused on human-robot interaction, behavioural responsiveness, and intelligent user engagement
- Worked on user-centric robot behaviours including emotional interaction, usability, and responsive interaction design
- Conducted prototyping, testing, and iterative improvements to optimise user experience and robot interaction workflows
- Winner – Robotics Category, UoA Part IV Project Display Day (2025)

Leadership & Volunteering

Treasurer & Event Coordinator – IEEE Student Branch | Jun 2023 – Nov 2025

- Organised and executed technical and community events
- Managed budgets and financial planning as Treasurer
- Demonstrated leadership, teamwork, and organisational skills

Marketing Officer – NZHSF | Feb 2024 – Nov 2024

- Created engaging marketing content and supported campaign initiatives
- Promoted student participation through communication strategies

Activities & Interests

- Active member of Women in Engineering (WEN) and UoA Basketball Club
- Passionate about drawing (anime and mandala art)
- Interest in animation, creative design, web development, and automotive technology platforms
- Enjoy reading and exploring new technologies

Reference

Available on request